University of Colombo, Sri Lanka

UCSC University of Colombo School of Computing

Bachelor of Science in Computer Science

First Year Examination — Semester I - UCSC AY21 [held in Sept./ Oct. 2024]

SCS 1306 — Linear Algebra

Two (2) Hours

103

Answer All Questions

Number of Pages = 12 Number of Questions = 4

To be completed by the candidate								
Index Number								

Important Instructions to candidates

- Students should answer in the medium of English language only using the space provided in this question paper.
- Note that questions appear on both sides of the paper. If a page or a part of this question paper is not printed, please inform the supervisor immediately.
- Write your index number CLEARLY on each and every page of this Question paper.
- This paper consists 3 structured questions and 10 MCQ based questions on **12** pages.
- Answer **ALL** questions.
- Write your index number on each and every page of the question paper.
- Do not tear off any part of this answer book. Under no circumstances may this book (or any part of this book), used or unused, be removed from the Examination Hall by a candidate.
- Calculators and any electronic device capable of storing and retrieving text including electronic dictionaries, smart watches and mobile phones are **not allowed**.

To be completed by the examiners

1	
2	
3	
4	
Total	

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1. (a). Let $A = \begin{pmatrix} 3 & 7 \\ 2 & 5 \end{pmatrix}$ and $B = \begin{pmatrix} 5 & -7 \\ -2 & 3 \end{pmatrix}$

Show that matrix A and matrix B are inverse of each other.

[5 marks]

(b). Consider the following linear system.

$$\begin{aligned} 2x + y - z &= 2 \\ 4x + 3y + 2z &= -3 \\ 6x - 5y + 3z &= -14 \end{aligned}$$

i. Write the augmented matrix of the above linear system.

[5 marks]

ii. Solve the above linear system using Gaussian Elimination with back substitution or Gauss - Jordan Elimination Technique.

[10 marks]

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(c). Consider the following 2 x 2 matrix.

$$C = \begin{pmatrix} 7 & 9 \\ 5 & 7 \end{pmatrix}$$

i. Compute the determinant of the matrix C.

[5 marks]

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ii. Find the inverse of the above 2 x 2 matrix C.

[5 marks]

2. Let $A = \begin{pmatrix} 6 & 2 & 0 \\ 2 & 3 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

(a). Determine the eigenvalues of matrix A

[6 marks]

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(b). Find the linearly independent eigenvectors corresponding to each eigenvalue obtained in part (a).

[12 marks]

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(c). If the matrix A is diagonalizable, determine the invertible matrix P and diagonal matrix D such that $P^{-1}AP = D$.

[4 marks]

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(d). If the $P^{-1} = \begin{pmatrix} \frac{2}{5} & \frac{1}{5} & 0 \\ \frac{1}{5} & -\frac{2}{5} & 0 \\ 0 & 0 & 1 \end{pmatrix}$, determine A^2.

[4 marks]

(e). Calculate the determinant (*det(A)*) and the trace (*tr(A)*) of the matrix A using the obtained eigenvalues.

[4 marks]

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3. The five corners of a house are given by the following column vectors.

$\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}1\\4\end{pmatrix}, \begin{pmatrix}3\\6\end{pmatrix}, \begin{pmatrix}5\\4\end{pmatrix}, \begin{pmatrix}5\\1\end{pmatrix}$

(a). Rotate the house 90° counterclockwise around the origin. Draw both the original figure (A) and the rotated figure (B) on a cartesian coordinate plane (with x and y axes). Write down the resulting coordinates.

[10 marks]

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(b). Scale the figure (B) by a factor of 2 along both axes. Determine the new coordinates of the five corners of the house.

[5 marks]

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(c). Reflect above scaled figure over the line $y = 0$ and determine the new coordinates of the shape.

[5 marks]

4. Question 4 has 10 MCQ based questions. **(2 marks x 10 = 20 marks)**

- Each MCQ question will have 5 (five) choices with one or more correct answers.
- Mark the correct answers in the provided space at the last page of this paper.
- Cross or color the **best suite answer** among the given options.
- There will be a penalty for incorrect responses to discourage guessing.

(a). Which of the following equation(s) is/are linear?

i. $3x + y - z = 5$
ii. $x = 7$
iii. $x^2 - 1 = 0$
iv. $sin(x) - z = 3$
v. $x + y - z + w = 5$

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(b). Determine the inner product (dot product) of the folowing two vectors: u and v.

$$u = \begin{pmatrix} -3 \\ 2 \\ 7 \end{pmatrix} \quad \text{and} \quad v = \begin{pmatrix} 9 \\ 2 \\ -5 \end{pmatrix}$$

i. -27	iii. 58	v. -35
ii. -58	iv. 35	

(c). Which of the following matrices are diagonalizable??

i. A matrix with all distinct eigenvalues
ii. A symmetric matrix
iii. A matrix that is not square
iv. A matrix with only one eigenvalue
v. A matrix with repeated eigenvalues but linearly independent eigenvectors

(d). Which of the following statements are TRUE about the determinant of a matrix?

i. The determinant of a singular matrix is 0.
ii. The determinant of an orthogonal matrix is always 1.
iii. The determinant of a diagonal matrix is the product of its diagonal elements.
iv. If the determinant of a matrix is 0, it has inverse.
v. If A and B are nxn matrices, then det(AB) = det(A).det(B)

(e). Which of the following(s) is/are TRUE about the Singular Value Decomposition (SVD)? (*U* and *V* are orthogonal matrices and Σ is a diagonal matrix)

i. $A = U\Sigma V^T$
ii. V is an inverse of U.
iii. Σ is always a square matrix.
iv. SVD can be used for dimensionality reduction.
v. SVD can be applied only for square matrices.

(f). Which of the following(s) is/are FALSE about the properties of the transpose of matrices? (*k* is a scalar, A and B are matrices.)

i. $(kA)^T = k^T A^T$	iii. $(A+B)^T = B^T + A^T$	v. $(AB)^T = B^T A^T$
ii. $(A^T)^T = A$	iv. $(A^T B)^T = B^T A$	

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(g). Which of the following statement(s) is/are TRUE about linear transformations?

i. Translation is a linear transformation.
ii. A linear transformation preserves vector addition and scalar multiplication
iii. A linear transformation always increases the dimension of a vector space.
iv. If a transformation T is linear, then $T(a+b) = T(a) + T(b)$ for all vectors a and b.
v. Every linear transformation does not map the zero vector to itself.

(h). Let $A = \begin{pmatrix} 1 & 5 & -2 \\ 3 & 7 & -9 \end{pmatrix}$, $B = \begin{pmatrix} -1 & 0 & 7 \\ -3 & 2 & 1 \\ -5 & 1 & 6 \end{pmatrix}$ and $C = \begin{pmatrix} 7 & -3 & 2 \\ 4 & 8 & 0 \\ -5 & 1 & 6 \end{pmatrix}$ Which of the following matrix operations is/are possible given the matrices A, B and C?

i. (A + B) + C
ii. A(B + C)
iii. BA + CA
iv. (B + C)A
v. ABC

(i). Find the transpose of the matrix $A = \begin{pmatrix} 7 & -3 & 2 \\ 4 & 8 & 0 \\ -5 & 1 & 6 \end{pmatrix}$

i. $\begin{pmatrix} 7 & -3 & 2 \\ 4 & 8 & 0 \\ -5 & 1 & 6 \end{pmatrix}$
ii. $\begin{pmatrix} 6 & 1 & -5 \\ 0 & 8 & 4 \\ 2 & -3 & 7 \end{pmatrix}$
iii. $\begin{pmatrix} 7 & 4 & -5 \\ -3 & 8 & 1 \\ 2 & 0 & 6 \end{pmatrix}$
iv. $\begin{pmatrix} 7 & 4 & 1 \\ -3 & 8 & -5 \\ 6 & 0 & 2 \end{pmatrix}$
v. $\begin{pmatrix} 7 & -5 & 6 \\ 4 & 8 & 1 \\ 2 & -3 & 0 \end{pmatrix}$

(j). Which of the following statement(s) is/are FALSE about a matrix??

i. Square matrices always have the same number of rows and columns.
ii. The inverse of a square matrix exists if its determinant is non-zero.
iii. A matrix can only have one row or one column.
iv. Matrix multiplication is always commutative.
v. All matrices are diagonal.

(a)	(i)	(ii)	(iii)	(iv)	(v)
(b)	(i)	(ii)	(iii)	(iv)	(v)
(c)	(i)	(ii)	(iii)	(iv)	(v)
(d)	(i)	(ii)	(iii)	(iv)	(v)
(e)	(i)	(ii)	(iii)	(iv)	(v)

(f)	(i)	(ii)	(iii)	(iv)	(v)
(g)	(i)	(ii)	(iii)	(iv)	(v)
(h)	(i)	(ii)	(iii)	(iv)	(v)
(i)	(i)	(ii)	(iii)	(iv)	(v)
(j)	(i)	(ii)	(iii)	(iv)	(v)

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